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1 import pandas as pd
2 from sklearn.preprocessing import LabelEncoder
3 from sklearn.neighbors import KNeighborsClassifier
4 from sklearn.model_selection import train_test_split
5 data = {
6     "weight": [150,170,120,200,180,130],
7     "color": ["red","orange","yellow","red","orange","yellow"],
8     "texture": ["smooth","rough","smooth","smooth","rough","smooth"],
9     "type": ["apple","orange","banana","apple","orange","banana"]
10 }
11 df = pd.DataFrame(data)
12 le_color = LabelEncoder()
13 le_texture = LabelEncoder()
14 le_type = LabelEncoder()
15 df["color"] = le_color.fit_transform(df["color"])
16 df["texture"] = le_texture.fit_transform(df["texture"])
17 df["type"] = le_type.fit_transform(df["type"])
18 X = df[["weight","color","texture"]]
19 y = df["type"]
20 X_train, X_test, y_train, y_test = train_test_split(*arrays: X,y,test_size=0.2)
21 knn = KNeighborsClassifier(n_neighbors=3)
22 knn.fit(X_train,y_train)
23 new_fruit = pd.DataFrame( data: [[160,2,1]], columns=["weight","color","texture"])
24 prediction = knn.predict(new_fruit)
25 print("Predicted Fruit:", le_type.inverse_transform(prediction))
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Predicted Fruit: ['banana']